SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA1021	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	Rashmitha Senthil Kumar	**Reg. No.:**	192525034
Date:	15/10/2026	**Duration:**	60 minutes

Code of Conduct

I __Rashmitha Senthil Kumar (192525034)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _Rashmitha .S_____

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Industry Context: Urban water distribution networks lose billions of gallons of clean drinking water annually due to hidden pipe bursts and slow sub-surface leaks. If an underground pipe burst goes unnoticed for hours, it can wash out roads flood basements, and waste expensive treated water. Municipal water teams need an automated network that tracks aqueduct pressure variations to pinpoint leak locations and isolate broken lines instantly.

.

Problem Statement: Design and develop a fault-tolerant Municipal Aqueduct Leak Detection Platform that monitors water pressure changes and coordinates valve systems across city grids. The platform must ingest high-velocity streaming data—including acoustic pipe sensors, water flow meters, valve statuses, and

soil moisture indicators—from thousands of utility nodes simultaneously. It must isolate breaks automatically.

Objectives:

1.Ingest and process telemetry from low-power aqueduct sensor networks with zero data loss.

2.Pinpoint underground leak positions using acoustic wave travel time difference checks.

3.Execute automated isolation commands to remote water valves when a pressure drop is verified.

4.Balance city water pressure targets dynamically to prevent pipes from cracking during demand drops.

5.Maintain unalterable, audit-ready records of city water utilization profiles for utility boards.

Expected Outcomes:

1.Significant reduction in clean water losses across urban distribution infrastructure.

2.Near-zero street flooding risks from large, uncontained underground water pipe bursts.

3.Lower maintenance costs via fast, automated pinpointing of fragile water lines.

4.Safe, steady water pressure delivery for city homes and commercial sectors.

FAULT-TOLERANT MUNICIPAL AQUEDUCT LEAK DETECTION PLATFORM

1. Identify the Project Vision and Stakeholders:

1.1 Project Vision:

Develop a dependable municipal water monitoring platform that transforms distributed sensor measurements into explainable leak alerts, location estimates, and verified valve actions. The platform should reduce avoidable water loss and response delays while preserving safe service conditions and accountable operational records.

1.2 Problem and Intended Value:

Hidden underground leaks can remain unnoticed while treated water escapes. Pressure variation alone cannot prove a burst because ordinary demand and pump operations also affect pressure. The platform must combine pressure, acoustic, flow, valve-status, and supporting soil-moisture evidence. Fault tolerance is necessary because communication and computing failures must not silently remove visibility or cause conflicting valve actions.

1.3 Project Objectives:

• Process simultaneous telemetry from thousands of registered utility nodes.

• Recover retained readings after temporary outages without duplicate utilization counts.

• Locate candidate leak sections using acoustic arrival-time differences.

• Automatically isolate confirmed breaks within approved engineering constraints.

• Coordinate bounded pressure adjustments during demand changes.

• Maintain tamper-evident utilization records and complete incident histories.

1.4 Scope and Planning Assumptions:

The product includes telemetry, asset registration, leak analysis, localization, valve coordination, pressure management, operator views, and audit reporting. Excavation, pipe repair, and drinking-water quality

certification are outside the software scope. Initial work uses simulated devices and valves; physical control follows engineering approval and pilot validation.

Zero loss is a recovery objective within configured retention limits, not an unconditional guarantee. Proposed timing, load, and accuracy thresholds in this document are acceptance targets for a controlled test environment, not measured results. No municipality-specific operating data or historical team velocity has been supplied.

1.5 Stakeholders and Roles:

Stakeholder	**Role and interest**
Utility board / sponsor	Approve funding, service priorities, policies, and acceptance targets; review outcomes.
Control-room operators	Monitor alerts, supervise automation, approve exceptions, and manage incidents.
Hydraulic engineers	Validate topology, pressure bounds, isolation rules, and safe actuation profiles.
Maintenance teams	Inspect estimated locations, calibrate devices, repair assets, and confirm findings.
Residents and businesses	Depend on stable supply and clear service interruption information.
Hospitals and priority users	Require protected-service rules and continuity arrangements.
IT and cybersecurity teams	Manage infrastructure, identities, keys, backups, monitoring, and recovery.
Device and valve suppliers	Provide supported interfaces, equipment limits, calibration, and servicing.
Auditors and oversight bodies	Review utilization totals, incident evidence, and authorized changes.

1.6 Scrum Responsibilities and Workshop:

A utility representative acts as Product Owner and orders the backlog. A Scrum Master facilitates planning and removes delivery obstacles. Developers collectively design, build, test, and document the increment. Engineers and operators are reviewers; they are not assumed to hold Scrum accountabilities unless assigned.

The proposed workshop starts with the vision and actor map, groups needs into epics, writes user stories, checks acceptance examples, orders dependencies, and estimates candidate stories. Developers then select a feasible Sprint 1 scope. Real names, availability, and estimates must be confirmed by the actual team.

2. Create Epics and User Stories

2.1 Epic Structure:

Epic	Outcome	Stories
E1 Trusted telemetry	Recognized devices and recoverable, validated observations.	US01-US03
E2 Operational visibility	Current readings and credible leak incidents.	US04-US05
E3 Acoustic localization	Reviewable candidate leak locations.	US06
E4 Safe isolation	Authorized valve actions and verified results.	US07-US08
E5 Pressure coordination	Bounded demand-responsive pressure control.	US09
E6 Continuity and accountability	Processing recovery, protected history, and utilization reporting.	US10-US12

2.2 User Stories:

US01 Device registration:

As a utility administrator, I want to register each device against a verified pipe asset, so that every reading has an accountable source.

US02 Validated ingestion:

As a control-room operator, I want valid sensor events to be stored with timestamps and quality flags, so that monitoring uses interpretable measurements.

US03 Outage recovery:

As an operations engineer, I want gateways to replay retained events after disconnection, so that recoverable history remains complete without duplicate totals.

US04 Live monitoring:

As a control-room operator, I want to view current readings and data freshness by zone, so that I can distinguish normal conditions from missing visibility.

US05 Correlated leak detection:

As a control-room operator, I want pressure and flow anomalies checked against acoustic evidence and known operations, so that likely leaks are distinguished from ordinary changes.

US06 Acoustic localization:

As a maintenance engineer, I want acoustic arrival-time differences mapped to candidate pipe sections with uncertainty, so that inspection can target a defensible area.

US07 Automatic isolation:

As a utility operator, I want confirmed breaks isolated automatically under approved safety rules, so that uncontrolled discharge is limited.

US08 Verified valve response:

As a utility operator, I want commanded and observed valve states compared, so that failed isolation is escalated rather than recorded as successful.

US09 Pressure coordination:

As a hydraulic engineer, I want pressure adjustments constrained by approved limits and rates, so that demand changes do not trigger unsafe corrections.

US10 Processing continuity:

As an operations engineer, I want another authorized processing instance to resume after failure, so that retained events continue to be processed.

US11 Protected incident history:

As an auditor, I want traceable incident and command records with integrity verification, so that unauthorized changes can be detected.

US12 Utilization reporting:

As a utility manager, I want utilization summaries traceable to accepted measurements and corrections, so that board reports can be explained and reconciled.

Each story describes an actor, capability, and benefit. Large control and recovery stories will be refined into smaller independently testable slices before selection. Access checks, evidence recording, and documented interfaces are cross-cutting requirements.

3. Define Acceptance Criteria

3.1 Given When Then Criteria for Every Story:

US01: Given an authorized administrator and a verified asset, when a unique device is registered, then its identity and asset link persist. When a duplicate ID or unauthorized request is submitted, then it is rejected and logged.

US02: Given 1,000 uniquely sequenced valid test events across all required event types, when ingested, then all are stored once with units, timestamps, and quality flags. Malformed events are quarantined with reasons; an unregistered source is rejected.

US03: Given 100 locally retained events during a simulated five-minute outage within buffer capacity, when connectivity returns and events replay twice, then exactly 100 unique events are stored. Given a sequence gap, then the missing range is reported.

US04: Given a reporting interval of five seconds, when no event arrives for three intervals, then the node is marked stale on the next five-second dashboard refresh. The last observation time remains visible.

US05: Given an approved labelled fixture set, when replayed, then all specified leak cases create incidents and specified normal-operation cases do not trigger isolation. Each decision identifies its evidence and rule version.

US06: Given a straight-pipe fixture with $L = 400$ m, $c = 1{,}000$ m/s, and $\Delta t = -0.10$ s, when localized, then the computed position is 150 m from A within 0.1 m numerical tolerance. Low-quality or physically inconsistent input yields an uncertain or rejected result.

US07: Given a confirmed simulated break and valid approved topology, when isolation conditions pass, then only permitted boundary valves receive commands with unique IDs and expiry times. Protected-service conflicts, expired authority, or invalid topology block automatic control.

US08: Given a simulated valve command and a configured ten-second acknowledgement timeout, when position feedback is absent, then the next monitoring cycle marks closure unverified and raises an alert. A repeated command ID causes no second actuation.

US09: Given approved pressure and adjustment-rate bounds, when demand drops, then proposed setpoints remain within both limits. Stale feedback or oscillation detected by the configured test rule suspends automatic adjustment and alerts the operator.

US10: Given accepted durable events and a running recovery instance, when the active processor fails, then processing resumes within the proposed 60-second test target. Replayed events do not create duplicate incidents or valve effects.

US11: Given a protected audit copy, when an entry is modified or removed, then integrity verification detects inconsistency. Unauthorized history edits are denied; authorized corrections append linked records.

US12: Given accepted flow records, quality flags, and a declared integration method, when a report is produced, then its totals match the reference fixture. Missing intervals are disclosed; estimates and corrections are labelled and traceable.

3.2 Interpretation of Acceptance Targets:

Numerical tolerances above verify software behaviour in simulation. They do not establish field localization accuracy, safe physical valve timing, or production recovery guarantees. Engineers and the Product Owner must approve operational thresholds before deployment.

4. Prioritize the Product Backlog

4.1 Ordering Method:

Use MoSCoW release classification together with a strict delivery order. Must denotes essential pilot capability; Should denotes important follow-on capability. Order considers service risk, business value, dependency readiness, and learning value. A later position does not imply that a safety feature is optional. Dependencies explain why trustworthy data comes before automatic control.

Rank / ID	Priority	Points	Dependencies	Value / reason
1 US01	Must	3	None	Establish trusted device and asset identity.
2 US02	Must	5	US01	Provide valid evidence to all later functions.
3 US03	Must	5	US02	Address recoverable data-loss risk early.
4 US04	Must	3	US02	Expose live status and stale measurements.
5 US05	Must	8	US02, US04	Detect credible incidents with explanations.
6 US06	Must	8	US05	Narrow field investigation to candidate sections.
7 US11	Must	5	US02; link US05	Protect evidence before control is enabled.
8 US08	Must	5	US01, US11	Validate valve feedback in simulation first.
9 US07	Must	8	US05, US06, US08, US11	Enable isolation only after prerequisites.
10 US10	Must	8	US03; integrated services	Verify end-to-end continuity before pilot acceptance.
11 US09	Should	8	US04, US08; approved model	Add constrained pressure coordination after control validation.
12 US12	Should	5	US02, US03, US11	Produce reconciled board-level summaries.

4.2 Dependency and Release Rationale:

Device identity precedes ingestion; ingestion precedes replay, monitoring, and analytics. Valve-response verification is developed against a simulator before automatic isolation. Baseline access control and event logging begin in Sprint 1; US11 expands these into integrity-protected incident evidence. Operational availability is a pilot release gate even though the integrated US10 test follows component development.

4.3 Definition of Ready:

A candidate story has a clear actor and benefit, testable criteria, known dependencies, available fixtures, a relative estimate, and no unresolved blocker. Stories touching physical control also require engineering constraints. A story that is too large is split before commitment.

4.4 Backlog Refinement:

The Product Owner revisits order with developers, operators, and engineers after each review. New risks and defects can change later work. Preserve the current Sprint Goal; renegotiate scope collaboratively when new evidence makes the original forecast unrealistic.

5. Plan the Sprint Backlog

5.1 Sprint Duration and Capacity:

Sprint 1 lasts two weeks, or ten working days. Planning assumption: three developers are available for six productive hours per day, giving 180 hours before allowances. Reserve 45 hours for Scrum events, coordination, and support, leaving 135 task hours. Selected tasks total 120 hours, leaving 15 hours of contingency. These are transparent planning assumptions, not confirmed staffing or historical velocity.

5.2 Selected Stories and Sprint Boundary:

Select US01 (3 points), US02 (5 points), and US03 (5 points): 13 story points. Build one demonstrable vertical slice from registered simulated devices through durable storage and replay. Leak localization and live valve operation are excluded from Sprint 1. A basic inspection view supports demonstration without claiming completion of the full US04 dashboard.

5.3 Actionable Development Tasks:

Task	Story	Work item	Hours
T01	US01	Define asset/device schema and unique identity rules.	8
T02	US01	Implement authorized registration and asset-link validation.	8
T03	US01	Test duplicates, invalid assets, and access denial.	8
T04	US02	Create fixtures for pressure, acoustic metadata, flow, valve status, and moisture.	10
T05	US02	Build ingestion, validation, persistence, and acknowledgement.	14
T06	US02	Test valid storage, rejected sources, and quarantine outcomes.	12
T07	US03	Implement bounded local buffering and reconnect replay.	12
T08	US03	Implement sequence deduplication and gap reconciliation.	12
T09	US03	Test outage, repeated replay, and missing-sequence cases.	10
T10	Shared	Integrate components and build minimal inspection view.	10

Task	Story	Work item	Hours
T11	Shared	Run grouped acceptance checks and resolve critical defects.	8
T12	Shared	Prepare setup guide, evidence, demonstration, and review notes.	8
Total		Task forecast; contingency outside this total.	120

5.4 Execution and Accountability:

Suggested sequence: days 1-2 establish contracts and fixtures; days 3-6 implement registration and ingestion; days 5-8 add replay; days 9-10 integrate, verify, and review. Developers self-manage assignments and pair-review critical changes. Daily Scrum exposes blockers. Task-hour estimates guide capacity; they are not a conversion formula for story points.

6. Estimate Effort Using Story Points

6.1 Estimation Technique:

Use Planning Poker with the Fibonacci scale 1, 2, 3, 5, 8, and 13. Developers estimate independently, reveal estimates together, discuss differences, and re-estimate. Points reflect relative complexity, integration effort, and uncertainty. They do not represent fixed hours. The values below are proposed estimates pending an actual team workshop.

6.2 Estimates and Technical Justification:

Story	Points	Justification
US01 Registration	3	Small schema and interface; identity uniqueness and access checks add moderate work.
US02 Ingestion	5	Multiple event types, validation, durable writes, acknowledgements, and quarantine.
US03 Replay	5	Buffer limits, retransmission, sequence reconciliation, and repeated-delivery tests.
US04 Monitoring	3	Read-only zone view and freshness indicators using established telemetry.
US05 Detection	8	Multi-source correlation, operational exceptions, fixture quality, and confidence.
US06 Localization	8	Timing interpretation, topology paths, uncertainty, and invalid-input handling.
US07 Isolation	8	Safety constraints, authority checks, protected users, and command lifecycle.
US08 Valve response	5	Feedback comparison, timeout handling, duplicate prevention, and simulated failures.

Story	Points	Justification
US09 Pressure	8	Bounded control, stability checks, hydraulic assumptions, and stale-data handling.
US10 Continuity	8	Failure injection, recovery coordination, replay, and duplicate-effect prevention.
US11 Audit	5	Append-only history, integrity verification, correction links, and restricted access.
US12 Reporting	5	Time integration, missing intervals, reconciliation, and traceable export.

6.3 Sprint 1 Estimate and Feasibility:

Sprint 1 total = US01 (3) + US02 (5) + US03 (5) = 13 points. Full proposed backlog = 71 points. Only the three selected stories form the Sprint 1 forecast. The team must validate its ability to deliver them against the 135-hour capacity assumption and 120-hour task forecast. No previous velocity is available, so 13 points must not be described as proven capacity.

6.4 Uncertainty and Scope Adjustment:

The main early uncertainties are device message formats, persistence behaviour, and replay semantics. Resolve them through simulated fixtures and a short technical investigation included in the relevant tasks. If integration exceeds capacity, developers and the Product Owner renegotiate scope while retaining a working recovery slice. Do not silently remove acceptance checks to declare completion.

Later 8-point stories require refinement before sprint selection. Split them by demonstrable behaviour, such as candidate localization before branched-network analysis, or permitted simulated isolation before field integration. Re-estimate after splitting instead of assuming the original total remains fixed.

7. Present the Sprint Goal and Expected Deliverables

7.1 Sprint Goal:

Demonstrate a trustworthy telemetry foundation that accepts readings from registered simulated aqueduct devices, retains valid measurements, and recovers buffered events after a temporary outage without duplicate records.

7.2 Expected Sprint Deliverables:

• Working device registration linked to verified test assets.

• Validated ingestion for all five required telemetry categories.

• Durable event storage, quality flags, and malformed-event quarantine.

• Gateway simulation with bounded buffering and reconnect replay.

• Sequence-based deduplication and visible missing-event ranges.

• Minimal inspection view plus setup and operator demonstration instructions.

• Acceptance evidence for US01-US03, a defect list, and review feedback.

7.3 Definition of Done:

A selected story is Done only when its acceptance criteria pass, changes are reviewed and integrated, required grouped tests pass, no unresolved critical defect remains, access checks and evidence recording are verified, and setup documentation is updated. The increment must run in the agreed test environment and be demonstrated. Unfinished stories return to the Product Backlog; partial work is not reported as completed points.

7.4 Sprint Review Demonstration:

Register a simulated device and show duplicate registration rejection. Send the 1,000-event fixture and inspect stored observations and quarantine records. Disconnect the gateway, retain 100 events, reconnect, and replay twice. Confirm exactly 100 unique recovered events and no duplicate utilization contribution. Introduce a sequence gap and show its reported range. Operators and engineers inspect the result and suggest backlog changes.

7.5 Test and Outcome Record:

Record	Status and evidence to retain
US01 acceptance	Not executed - retain registration, duplicate, and access-denial results.
US02 acceptance	Not executed - retain fixture counts, stored records, and rejection reasons.
US03 acceptance	Not executed - retain outage timeline, sequence reconciliation, and duplicate counts.
Defect and correction log	Record issue ID, expected/actual behaviour, severity, correction, and retest outcome.
Review and retrospective	Record feedback, capacity differences, agreed improvements, and next backlog order.

7.6 Conclusion:

The workshop converts the aqueduct problem into six epics and twelve traceable user stories with measurable acceptance criteria. Dependency-based ordering places trustworthy data before high-impact control. Sprint 1 delivers a realistic simulated ingestion-and-recovery foundation; later increments add detection, localization, verified isolation, continuity, pressure coordination, and reporting. Operational deployment remains subject to engineering approval and measured pilot validation.